LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Daily Practice: Ch 11 Electricity

Chapter / Topic: Electricity · Week 6 of the 90-day plan

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DAY 1: CURRENT, POTENTIAL DIFFERENCE AND OHM'S LAW

Q1. One ampere is equal to:

Answer: (a) 1 coulomb per second (1 J/s is a watt; 1 J/C is a volt)

Q2. At a constant temperature, the resistance of a wire does NOT depend on:

Answer: (b) the potential difference applied across it: if V is doubled, I doubles and R stays the same

Q3. A current of 2 A flows through a wire for 3 minutes. The charge that flows is:

Answer: (c) 360 C: Q = It = 2 × 180

Q4. The potential difference between two points is the:

Answer: (b) work done per unit charge: V = W/Q

Q5. Draw the circuit symbols for: a cell, a battery, a closed plug key, a resistor, a rheostat (variable resistor), an ammeter and a voltmeter.

Answer: Cell: one long thin line (+) and one short thick line (−). Battery: two or more cells joined. Closed plug key: a key symbol with a dot (plug) in it. Resistor: a zig-zag line. Rheostat: a zig-zag with an arrow across it. Ammeter: a circle with A. Voltmeter: a circle with V.

Q6. A potential difference of 6 V is applied across a 24 Ω resistor. Find the current. What will the current be if the potential difference is doubled?

Answer: I = V/R = 6/24 = 0.25 A. If V is doubled to 12 V, I = 12/24 = 0.5 A: the current also doubles (Ohm's law).

Q7. Wire P has length L and area of cross-section A. Wire Q, made of the same material, has length 2L and area of cross-section 4A. Compare their…

Answer: $R_P = \rho L/A$. $R_Q = \rho(2L)/(4A) = ½ \times \rho L/A$. So the resistance of Q is half that of P.

Q8. A nichrome wire 2 m long, with an area of cross-section of 0.5 mm^2, has a resistance of 4 Ω. Find the resistivity of nichrome.

Answer: A = 0.5 mm^2 = 0.5 × 10^{-6} m^2.
ρ = RA/L = (4 × 0.5 × 10^{-6}) ÷ 2 = 1 × 10^{-6} Ω m.

DAY 2: RESISTORS IN SERIES AND PARALLEL

Q9. Resistors of 3 Ω, 6 Ω and 9 Ω are connected in series. The equivalent resistance is:

Answer: (c) 18 Ω: in series, R = 3 + 6 + 9

Q10. Resistors of 6 Ω and 3 Ω are connected in parallel. The equivalent resistance is:

Answer: (a) 2 Ω: 1/R = 1/6 + 1/3 = 1/2

Q11. In a parallel combination of resistors, which quantity is the same for every resistor?

Answer: (b) the potential difference

Q12. You have three 6 Ω resistors. How will you connect them to get an equivalent resistance of (a) 9 Ω, (b) 4 Ω?

Answer: (a) Two in parallel (3 Ω), then that pair in series with the third: 3 + 6 = 9 Ω.
(b) Two in series (12 Ω), then that pair in parallel with the third: (12 × 6)/(12 + 6) = 4 Ω.

Q13. Resistors of 5 Ω, 10 Ω and 15 Ω are connected in series to a 6 V battery. Find the current and the potential difference across each resistor.

Answer: Total R = 30 Ω, so I = 6/30 = 0.2 A.
$V_5 = 0.2 \times 5 = 1$ V, $V_{10} = 0.2 \times 10 = 2$ V, $V_{15} = 0.2 \times 15 = 3$ V. (Check: 1 + 2 + 3 = 6 V.)

Q14. Resistors of 10 Ω and 40 Ω are connected in parallel, and this combination is connected in series with a 2 Ω resistor and a 10 V battery. Find (i)…

Answer: (i) Parallel part: (10 × 40)/(10 + 40) = 8 Ω. Total = 8 + 2 = 10 Ω.
(ii) I = 10/10 = 1 A.
(iii) Potential difference across the parallel part = 1 × 8 = 8 V. Current through 10 Ω = 8/10 = 0.8 A; through 40 Ω = 8/40 = 0.2 A. (Check: 0.8 + 0.2 = 1 A.)

Q15. Which has the higher resistance: a 100 W bulb or a 60 W bulb, both rated for 220 V? Show by calculation.

Answer: $R = V^2/P$. 100 W bulb: 220 × 220/100 = 484 Ω. 60 W bulb: 220 × 220/60 ≈ 807 Ω. The 60 W bulb has the higher resistance.

DAY 3: HEATING EFFECT AND ELECTRIC POWER

Q16. **Assertion (A):** Electric power is measured in kilowatt-hours. **Reason (R):** 1 kWh = 3.6×10^6 J.

Answer: (d) A is false: the kWh is a unit of energy; power is measured in watts. R is true.

Q17. The heat produced in a resistor is $H = I^2Rt$. If the current is doubled, the heat produced in the same time becomes:

Answer: (d) four times, because heat depends on the square of the current

Q18. In a circuit, a fuse is connected:

Answer: (a) in series with the device, so that it breaks the circuit when the current is too high

Q19. An electric kettle is rated 880 W, 220 V.

(i) I = P/V = 880/220 = 4 A.
(ii) R = V/I = 220/4 = 55 Ω.
(iii) The 5 A fuse. It is a little above the normal current of 4 A, so it will not blow in normal use but will melt if the current rises dangerously. A 2 A fuse would blow at once; a 15 A fuse would not protect the kettle.

Q20. (a) Starting from P = VI, show that $P = I^2R$ and $P = V^2/R$. (b) A motor draws a current of 3 A from a 220 V supply. Find its power, the energy it uses…

Answer: (a) Using V = IR: P = VI = (IR)I = I^2R. Using I = V/R: P = V(V/R) = V^2/R.
(b) P = VI = 220 × 3 = 660 W = 0.66 kW. Energy = 0.66 × 5 = 3.3 kWh (3.3 units). Cost = 3.3 × 6 = ₹19.80.

Q21. The 100 W and 60 W bulbs from Day 2 are connected in series to a 220 V supply. Which one glows brighter? Explain.

Answer: The 60 W bulb. In series, the same current flows through both, so the power used ($P = I^2R$) is greater in the bulb with the higher resistance. The 60 W bulb has the higher resistance (about 807 Ω against 484 Ω), so it glows brighter.